Motor temperature calibration: rider checklist

Your steps only, from the full plan (docs/motor_temperature_calibration_plan.pdf). Claude sends every command. Where it says **tell Claude**, nothing is sent until you confirm in chat. **Updated 7 Nov 2026.** Phase 1 is done (resistance 56.3–56.6 mΩ, pass) and the bench session fitted the model to 1.1 °C, so the 40-minute cooldown is no longer needed. The ride now only measures cooling at speed and checks the model.

Every measurement gives a small jerk of the rear wheel. When a step says wheel up: rear wheel off the ground (or rear brake held firmly), and keep hands and feet clear of the wheel.

1. Before the first ride on the new script

about 5 min

- ☐ Both wheels in the air: pull the brake lever while spinning the wheels gently, check the ABS (wheels must not lock).
- ☐ Rear stick test: rear wheel down, brake held, gentle throttle. Nothing unusual.
- ☐ First few minutes slow. The battery limit is now 110 A total: if the scooter cuts out under hard acceleration (BMS), tell Claude.
- ☐ Outdoors with GPS fix: the clock on the screen shows the right local time.

2. Morning of the ride, before riding

phase 2 · about 5 min

- ☐ Scooter parked indoors for at least 4 hours (overnight is best).
- ☐ Switch it on. Read the room temperature on a household thermometer: _____ °C
- ☐ Rear wheel up (or brake held). **Tell Claude** the room temperature and confirm. Claude stores the cold reference.
- ☐ Laptop helper window open, scooter on Wi-Fi.

3. First 10 minutes of the ride

phase 3

- ☐ Ride normally.
- ☐ Stop 3–4 times: standing still, throttle released, hold the brake lever for 3 seconds. A small jerk = a measurement.

4. The climb

phase 4

- ☐ Hardest climb available (the evening route, or a steep hill 2–3 times), full throttle where safe.
- ☐ Every 2–3 minutes: stop, stand still, hold the brake for 3 seconds.
- ☐ Aim: rear motor at 80–100 °C on the screen.

Stop climbing if the motor temperature on the screen passes 110 °C (it shows the model) and ride gently. The current reduction starts by itself at 100 °C.

5. Ride home: cooling at speed

phase 5

- ☐ Ride on the flat at a steady speed, ideally two stretches: one at about 25 km/h and one at about 40 km/h.
- ☐ Every 3–4 minutes: stop, hold the brake for 3 seconds, ride on. Keep the stops short (under 30 seconds).
- ☐ Do not switch the scooter off until you are home.

Afterwards: tell Claude you are done. Claude pulls the logs, fits the cooling at speed, checks the model against the measurements and, if it holds, switches the measurement correction on.